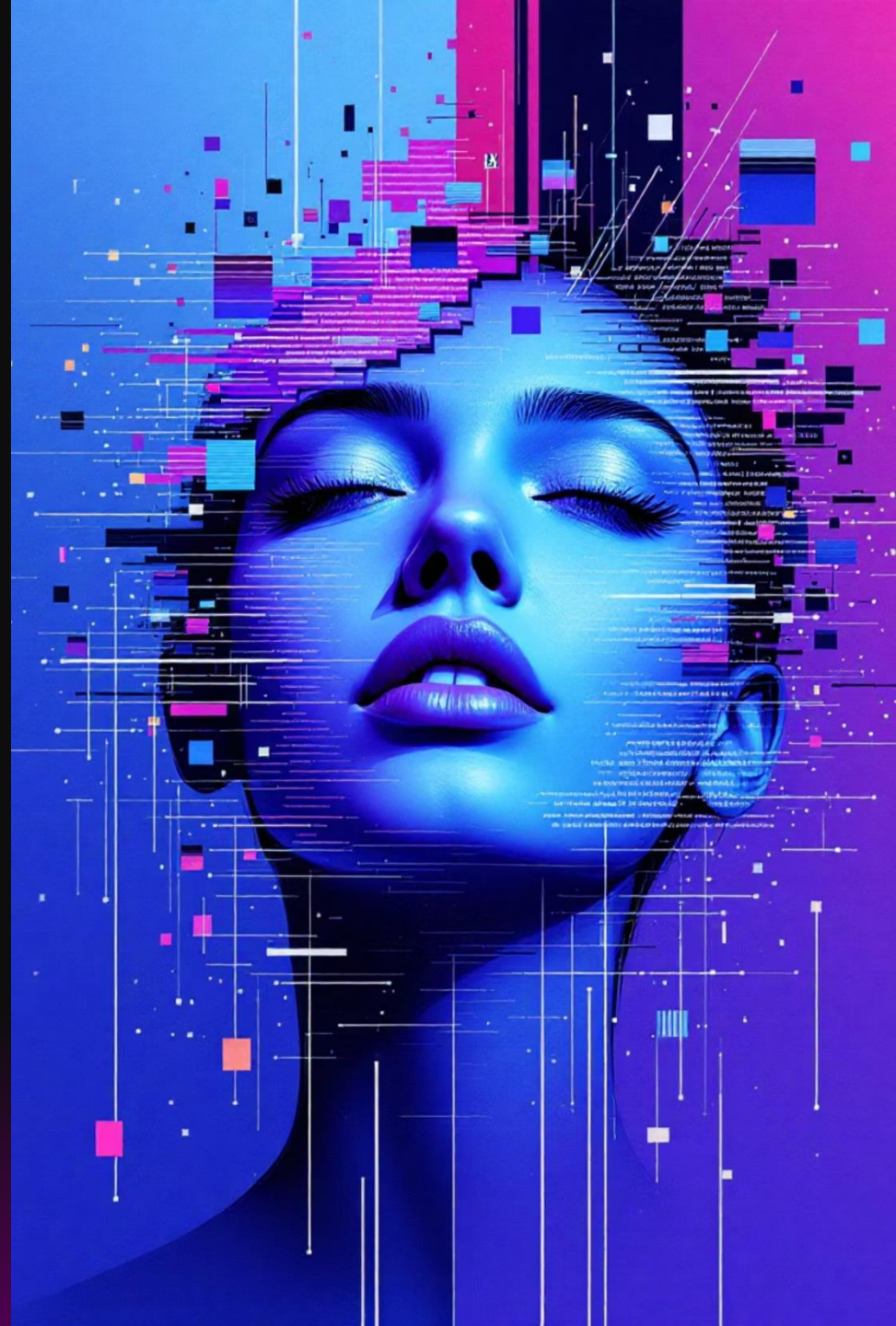
Combating Deepfake - Driven Fraud and Misinformation

Identity Theft: Deepfakes now bypass biometric Know Your Customer (KYC) procedures and enable the impersonation of senior executives.

Societal Threat: Used extensively to spread political misinformation, manipulate public opinion, and orchestrate social engineering attacks.

Financial Impact: Global losses due to deepfake-enabled fraud exceed hundreds of millions of USD annually.

Our AI/Machine Learning (ML) project introduces a robust detection pipeline designed to identify and prevent deepfake videos across critical sectors: finance, politics, and human-trafficking investigations.



Research & Analysis: The Escalating Deepfake Threat

The sophistication of synthetic media poses a critical, evolving threat that transcends traditional security measures. Modern deepfakes are designed to exploit subtle vulnerabilities.

$25.6M

Largest Loss Reported

Amount lost in a single deepfake video call scam targeting a London-based multinational firm (2024).

5 min

Frequency of Attempts

Estimated frequency of deepfake fraud attempts occurring globally in 2024, highlighting the volume of attacks.

Bypassing Liveness

Attackers use AI face swaps and virtual camera spoofing to circumvent standard liveness detection protocols.

Subtle Artefacts

Key challenges include detecting subtle manipulation like facial morphing, inconsistent lip-sync, and AI-generated non-human faces.

Process Model & Architecture

Due to restricted access to the Hadoop cluster, Spark ingestion was executed in a standalone mode. The pipeline demonstrates the same scalable logic, reading, balancing, and manifest generation as would occur in a cluster environment, thereby fulfilling the Big Data application objective.

The project used a two-tier data storage strategy:

- Cropped faces were first saved as JPEGs to allow visual inspection and reusability across experiments.
- CNN-extracted 256-D embeddings stored in .npz format to support fast, lossless numerical access for LSTM sequence modelling with a resource aware choice balancing storage, performance, and hardware limits (CPU 16 GB RAM, Colab L4 GPU).

During experimentation, both EfficientNet B2 and Xception were evaluated for CNN-based for spatial detection. Xception was selected as the final backbone with balanced performance between real and fake class recall, better handling of subtle facial texture inconsistencies, and more stable training convergence on limited GPU resources.

Performance is measured using Accuracy, Precision, Recall, and F1-score, with threshold tuning for high sensitivity in KYC scenarios.

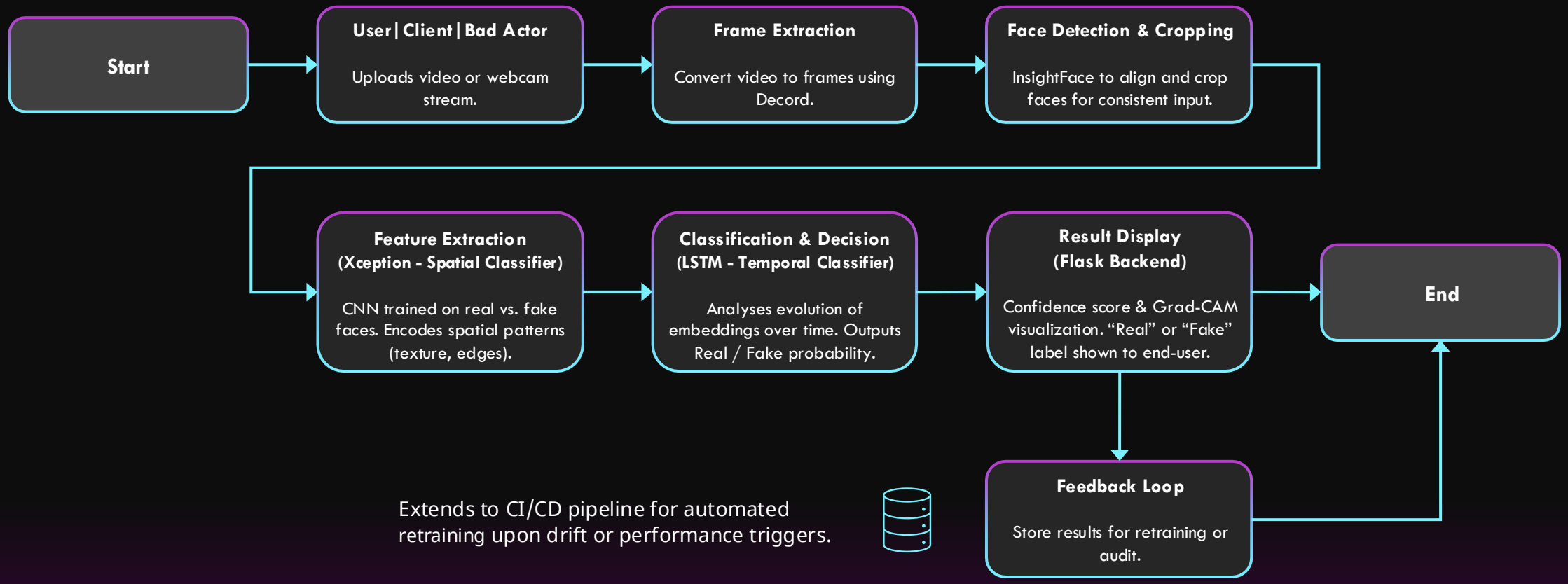
A Long Short-Term Memory (LSTM) layer analyses the sequence of embeddings to detect motion inconsistencies (e.g., blinking, lip-sync errors).

Initial prototype deployment via a Flask web portal (upload). Future scaling involves Kafka/Spark Streaming integration for real-time KYC alerts at scale (web-cam)

This six-stage architecture from Spark ingestion to real-time (future) deployment demonstrates both Big-Data scalability and Deep-Learning application.

1. **Data Ingestion (Big Data Layer)**
   - Apache Spark ingest 2000 videos (1000 real/ 1000 fake)
   - Balanced Train/Val/Test splits via Spark manifest generation – no HDFS used.
2. **Pre-processing (Data Engineering)**
   - Frame extraction with Decord and face alignment with Insightface.
   - Cropped faces stored as (299,299) JPEGs.
3. **Feature Extraction (AI Layer)**
   - Xception CNN (pretrained ImageNet) – 256 D embeddings stored as .npz for fast I/O
4. **Temporal Modeling (AI Layer)**
   - LSTM captures motion (lip sync, blink, head movement).
5. **Evaluation & Optimization (AI Layer)**
   - Metrics; Accuracy, Precision, Recall, F1; threshold tuning for KYC sensitivity.
6. **Deployment (Application Layer)**
   - Flask web portal for upload/webcam feed.
   - Future – Kafka / Spark Streaming for real-time KYC.

End-to-End Process from Video Upload to Real/Fake Classification



Extends to CI/CD pipeline for automated retraining upon drift or performance triggers.

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Prototype Demonstration & Results Preview

The current prototype provides immediate, actionable feedback on video authenticity and manipulation indicators.

1

Analysis Modes

Supports both offline analysis via video file upload and real-time analysis through live-stream/webcam feed processing.

2

Confidence Scores

The system generates clear confidence scores indicating the probability of a video being a deepfake vs. authentic.

3

Visual Indicators

Real-time visual overlays highlight specific facial artefact regions where manipulation is detected, enhancing transparency.

Initial Baseline Accuracy

~81%

Achieved using the Xception CNN model only. LSTM integration is currently underway. Future trails on Vision Transformer (VIT).

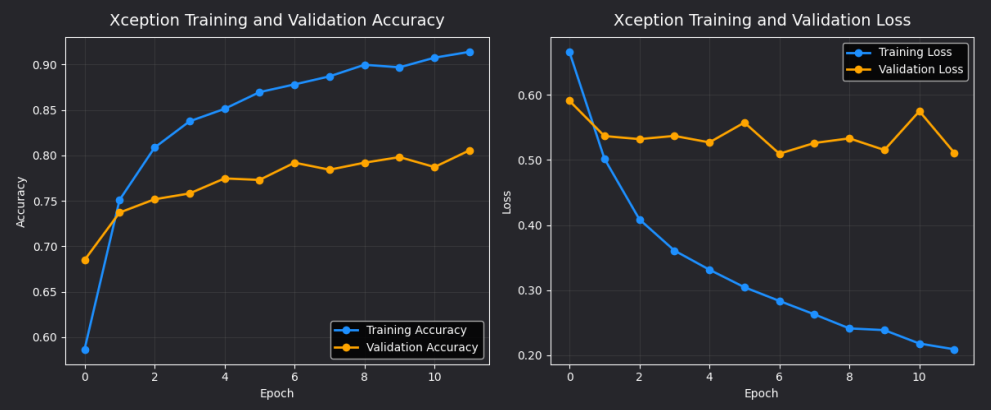
Target Accuracy (Temporal Fusion)

>90%

Our goal for the final integrated system, combining spatial and temporal features, is to exceed the 90% threshold.

Prototype Demonstration & Results Preview

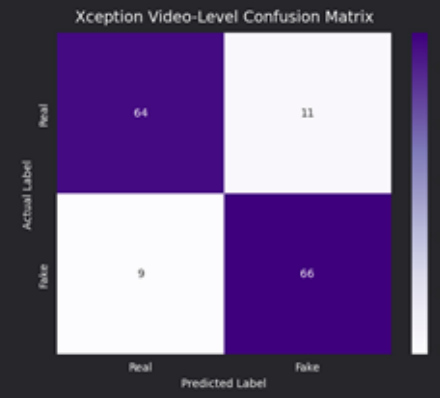
Training Dynamics



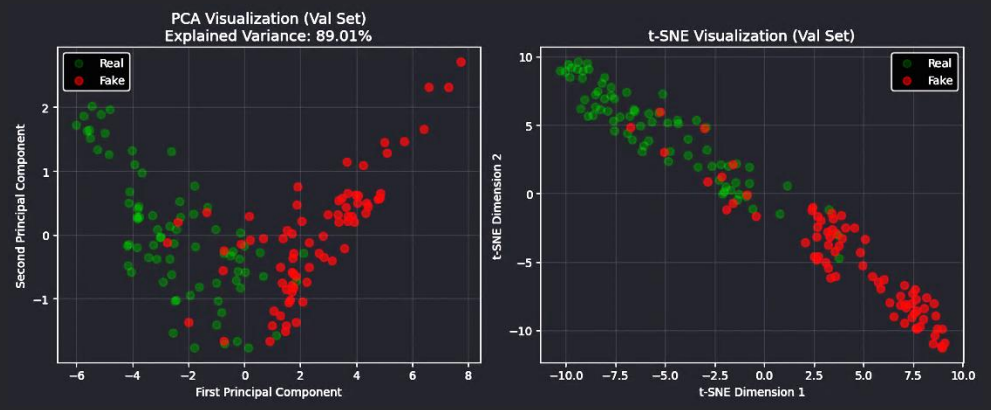
Xception Classification Report – Video Level

Class	Precision	Recall	F1-Score	Support
Real	0.8767	0.8533	0.8649	75
Fake	0.8571	0.8800	0.8684	75
Accuracy			**0.8667**	150
Macro Avg	0.8669	0.8667	0.8666	150
Weighted Avg	0.8669	0.8667	0.8666	150

Confusion Matrix

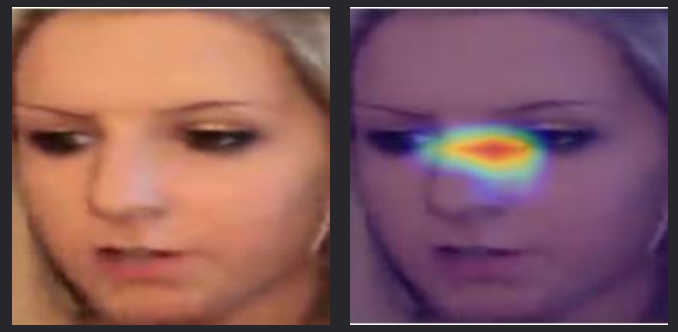


Class Separability in Latent Space



Cropped Face Examples

Grad-CAM (Detected Artefacts) Examples



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